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Chapter 23
Adding Uncertainty: Risk of Ruin with Uncertain Win Rates

One of the authors has a recurring vision of a worдd where tiny holographic projections appear above each player's head at the table, showing that player's precise "true" expectation in the current hand/game/session, including the players in the game, the player's state of mind, and so on. Unfortunately, such holograms do not exist, and so we must estimate and infer. In Part I, we discussed estimating parameters. In this section, we look at combining the risk of ruin formula we calculated in the previous chapter, which rely on a solid, known win rate, with an uncertainty distribution of win rates inferred from observed data. Throughout this section, we focus on limit poker, although these principles apply equally to no-limit ring games. However, since this analysis relies on the normal distribution risk of ruin formula, it cannot be used as is to solve problems involving tournaments.

Consider a player who has played a number of hours N with a win rate of *w* and a standard deviation of *s*. We can observe first of all that sample standard deviations converge on the population standard deviation quite quickly. So throughout this analysis, we will use *s* as the standard deviation σ of the game. However, we know from our experience that *w* does not converge on μ the player's "true" win rate, very quickly at all because the magnitude of *s* is high relative to *w*.

To see this, imagine that we knew μ=1 bet/hr and σ = 12 bets/hr and we sampled 100 hours and called the result of that sampling *w*. The result *w* of this sample would be 68% likely to fall between 0 and 200 bets; 32% of the time, the result would be more than 1 bet/hr away from the mean. Let's say this player has a current bankroll of200 bets. The difference between the risk of ruin at 0 bets/hr and that at 2 BB/hr is quite large!

0 bets:

$R(200) = 1$

2 bets:

$R(200) = e^{-800/144} = 0.387\%$

So this is a nearly 100% range of RoR values, all within one standard deviation of the population mean. There is even a fairly substantial difference between RoR at win rates of 0.8 BB/hr and 1 BB/hr:

0.8 bets:

$R(200) = e^{-320/144} = 10.84\%$

1 bet:

$R(200) = e^{-400/144} = 6.22\%$

However, the process of acquiring enough empirical data to tell the difference between the former and the latter is an arduous task. So we have a dilemma. We can use our win rate and standard deviation *w* and *s* and simply assume that they are accurate predictors; however, our true population mean may differ significantly from the observed values. A more robust approach is to account for the uncertainty of our win rate by using a distribution of win rates in lieu of

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assigning a single win rate. We can do this by utilizing a normal distribution with a mean of our observed win rate and a standard deviation equal to the standard error of the observed win rate. Then we can utilize the normal distribution risk of ruin function as a value function and this hypothesized distribution of win rates as the corresponding density function. Then the expected value of the risk of ruin function will be the integral of the product of those two functions.

It seems valuable to take a moment to try to explain this process in somewhat less technical language. We have some observed win rate *w*. So We are saying that we can make a normal distribution curve around *w*. The closer we are to *w*, the higher the probability is that we are at our true win rate. At each *w*, we can use the risk of ruin function we've calculated all along to find the risk of rum for that particular win rate. Then if we multiply all the risk of ruin values by the probability that the value is our true win rate and sum it all up, we get a "weighted" risk of ruin value that is more accurate than simply guessing at a particular win rate.

What's important to note here is that we can do this with only a little data about the game. If we only have a small number of hours, the dispersion of the distribution of win rates will be quite large. As we will see, the dispersion of the distribution of win rates is quite a critical value for finding this weighted risk of ruin.

The reader may also remember that we warned against using simply applying a normal distribution to observed results when we spoke of confidence intervals in Part I. This warning is still in effect; in fact, using this method doesn't incorporate information about the underlying distribution. This is particularly true where the observed mean is very high, when Bayesian analysis would dampen the observation relative to the population's win rate distribution. However, as this bias is restricted to the right tail of the normal distribution, its magnitude is fairly small compared to the gain in accuracy we get from applying the method. And attempting to correct this bias requires that we make attempts to characterize the shape of the *a priori* distribution from which this player is selected.

With that in mind, let us specify the terms of the model.

A player has played *N* hands with win rate *w* and standard deviation per hand *s*. The player has a bankroll *b*. His risk of ruin for any particular win rate Is:

$R(w,b) = \exp\left(\frac{-2wb}{s^2}\right)$

$f(x) = \exp\left(\frac{-2wb}{s^2}\right)$

The standard error of his win rate observation is:

$$\sigma_w = s/\sqrt{N}$$

His observations about win rate create a normal density function around mean *w* and standard deviation *σw*.

$$p(x) = \left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)exp\left(\frac{-(x-w)^2}{2{\sigma_w}^2}\right)$$

Using Equation 22.3, we have:

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$$< v(x) >= \int_{-\infty}^{\infty} v(x)p(x)dx$$

This equation calls for us to integrate from -∞ to +∞; however, we know that the risk of ruin for win rates to the left of zero is always 1. Hence, we can split this integral into two parts: the integral from 0 to ∞ of the value function (risk of ruin) multiplied by the density function (normal for the distribution of win rates) plus the density function from -∞ to 0. This last equals the probability that the win rate is equal to zero or less.

$$< v(x) >= \left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)\int_0^{\infty} exp\left(\frac{-2xb}{\sigma^2}\right)exp\left(\frac{-(x-w)^2}{2{\sigma_w}^2}\right) + p(w<0)$$

The last term is the most important feature of this approach to the method. What it captures is that the uncertainty surrounding the win rate and the idea that having a bankroll of any size does not protect against the risk of ruin that comes from being a losing player. This probability, as we shall see shortly, is directly tied to the uncertainty σw; the higher this dispersion is, the greater the chance that the win rate is below zero. Also, the risk of ruin for win rates that are positive but very low is also quite high itself.

We can also now point to a very practical reason for using the normal distribution for the distribution of win rates. Because both the normal distribution and the risk of ruin formulas are exponentials, the above expression simplifies!

$$\left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)\int_0^{\infty} exp\left(\frac{-2xb}{\sigma^2}\right)exp\left(\frac{-(x-w)^2}{2{\sigma_w}^2}\right)dx + p(w<0)$$

$$\left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)\int_0^{\infty} exp\left(\frac{-2xb}{\sigma^2} - \frac{x^2-2xw+w^2}{2{\sigma_w}^2}\right)dx + p(w<0)$$

$$\left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)\int_0^{\infty} exp\left(-\frac{x^2-2x\left(2b\frac{\sigma_w^2}{\sigma^2}-w\right)+w^2}{2{\sigma_w}^2}\right)dx + p(w<0)$$

Now call $u = 2b\frac{\sigma_w^2}{\sigma^2} - w$

$$\left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)\int_0^{\infty} exp\left(-\frac{(x+u)^2+w^2-u^2}{2{\sigma_w}^2}\right)dx + p(w<0)$$

$$\left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)\int_0^{\infty} \exp\left(-\frac{(x-(-u))^2}{2{\sigma_w}^2}\right)exp\left(-\frac{w^2-u^2}{2{\sigma_w}^2}\right)dx + p(w<0)$$

That second exponential doesn't contain any x values, so we can pull it out of the integral, leaving us with a constant multiplied by a normal distribution function:

$$exp\left(-\frac{w^2-u^2}{2{\sigma_w}^2}\right)\left(\frac{1}{\sigma_w\sqrt{2\pi}}\right)\int_0^{\infty} \exp\left(-\frac{(x-(-u))^2}{2{\sigma_w}^2}\right)dx + p(w<0)$$

Now we need to introduce a little bit of notation and discuss normal distributions a bit.

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Recall that in Chapter 3 we defined:

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} \exp\left(-\frac{x^2}{2}\right)$$

where z was a normalized z-score.

Φ(t) represents the area under the normal distribution curve for a distribution with a mean of 0 and a standard deviation of 1 to the left of *x*, and we call it the ***cumulative normal distribution function.***

If this normal distribution function had integrands of -∞ and ∞, then it would equal one. However, it is integrated from 0 to ∞. What this means is that it equals the cumulative normal distribution function for a z-score of *–u/σw*, or $(w - 2b\frac{\sigma_w^2}{\sigma^2})/\sigma w$. This type of value can be looked up in a table or retrieved using a spreadsheet function.

We are left with:

$$exp\left(-\frac{w^2 - u^2}{2{\sigma_w}^2}\right)(\Phi(-u/\sigma w)) + p(w < 0)$$

$$-\frac{w^2 - u^2}{2{\sigma_w}^2} = -\frac{w^2 - (4b^2\frac{\sigma_w^4}{\sigma^4} - 4bw\frac{\sigma_w^2}{\sigma^2} + w^2)}{2{\sigma_w}^2} = (2b^2\frac{\sigma_w^2}{\sigma^4} - \frac{2bw}{\sigma^2})$$

$$exp(2b^2\frac{\sigma_w^2}{\sigma^4} - \frac{2bw}{\sigma^2})(\Phi(-u/\sigma w)) + p(w < 0)$$

We have one additional simplification:

$$RoRU = R(w,b)exp(2b^2\frac{\sigma_w^2}{\sigma^4})(\Phi((w - 2b\frac{\sigma_w^2}{\sigma^2})/\sigma_w)) + (-w/\sigma_w) \qquad \textbf{(23.1)}$$

This is the formula for risk of ruin under uncertainty (RoRU). It contains what are essentially four separate items.

Tire first is *R(w,b),* which is simply the normally distributed risk of ruin for the base observed win rate. This factor is then multiplied by the second element, an exponential involving the square of bankroll and the standard deviations of both the game and the win rate observations. All of this is then multiplied by a factor of the cumulative normal distribution for an additional expression involving *w*, the bankroll *b*, and the standard deviations of both distributions. This product (of the first three terms) is the weighted risk of ruin for the player for win rates that are positive. The last term is the cumulative standard normal distribution for win rates that are negative, multiplied by the risk of ruin for those win rates, which is 1.

Here's a table which looks at some risk of ruin calculations, using the normal distribution RoR and also this RoRU (risk of ruin under uncertainty) calculation:

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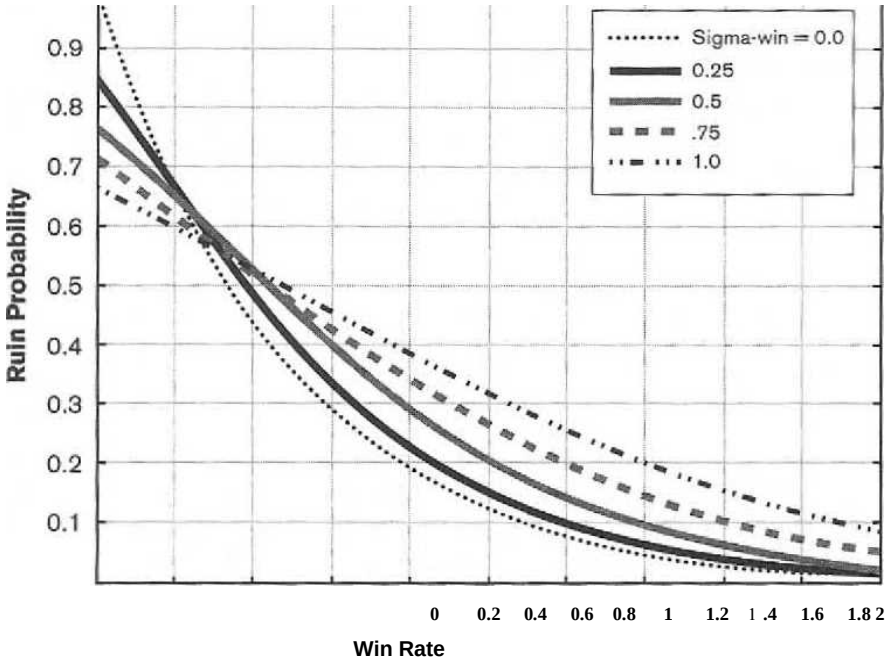

Figure 23.1 RoRU vs win rate (bankroll -100, std. dev. -10 bets)

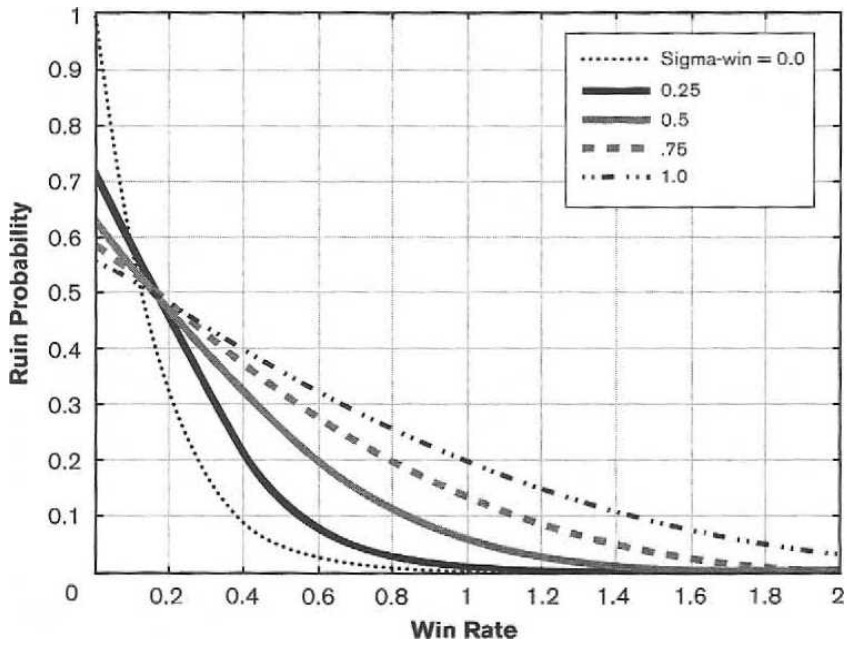

Figure 23.2 RoRU vs win rate (bankroll + 300 bets, std. dev. = 10 bets)

Of note in these figures is that when the uncertainty of win rate is high (the sample size is small), RoRU predicts a less than 100°/o chance of going broke, even if the win rate is negative up to this point. This is because the sample might just have been unlucky, and the true win rate might be above zero, in which case the RoR would be less than one,

RoRU is a powerful method of incorporating uncertainty about win rates into our risk of ruin calculations; we can adjust our expectations and understanding of the risk of going broke and include a quantitative adjustment for the possibility that we have just gotten somewhat lucky so far.

The risk of ruin under uncertainty formula we just derived is more complex, of course, than the previous formulas. The characteristic that makes it extremely useful, however, is that it attempts

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to account in some way for the tendency for players to use win rates that are too high in estimating their risk of ruin. In terms of staying in action, it is difficult to overestimate the importance of this tendency and the resultant loss of bankroll principal that results.

Bankroll Considerations

We believe that most players overestimate then win rates for risk of ruin calculations for any of a number of reasons.

1) Initial Success - Players who begin then poker careers with a series of wins are more likely to continue playing. Players who begin their poker careers with a series of losses are far more likely to simply quit the game. This effect has perhaps a small selection effect toward players with higher EVs, but this effect is swamped by variance.

If we look at the set of all players who begin then careers by winning, we will find quite a number who are relatively poor players who have simply had positive variance events. These players will likely rely on their past (inflated) win rates as predictors of their future (expected) win rates and thus overestimate their prospects for the future. This effect is particularly pronounced for players who are considering playing professionally. The most likely time for a player to consider playing full-time is after some sort of upswing. Such upswings, while they may be real, are often positive statistical swings that may not reflect an underlying change in expectation.

2) Ego - Poker is an ego-driven game. Many players are convinced that they are much stronger players than they really are, that their results up to now have been due primarily to bad luck and not their poor play. This, of course, is sometimes true; but more often, it may be a combination of the two.

3) Misinformation - There is a tremendous amount of information floating around the poker community, both from public forums such as rec.gambling.poker, other Internet forums, magazines, and now even mainstream media sources. In these media, players can claim any (inflated) win rate they choose without any realistic chance of being exposed. This leads to a general inflation of "win rate stories," as individuals do not wish to feel inferior to outer players.
In addition, the poker literature has a disturbing tendency to dismiss questions of what constitutes realistic expectation in different types of games, preferring to rely on the "1 big bet per hour" platitude as a guide to what a typical winning player can expect.

4) Changing Game Conditions - Players try to project their win rates from now into the future. However, game conditions change - games dry up or shift to a different preferred game, as lowball games did with the introduction of holdem. Likewise, players may improve over time. Many things can change that make our estimates of the future inaccurate.

Very few poker players, especially those who have spent time and energy trying to improve, are willing to admit that there is a significant chance that they are losing players overall. In all honesty, overestirnating one's win rate is very close to a universal property of poker players. There is one additional reason that win rates for risk of ruin calculations are often overstated, and it has to do with the fact that most poker players do not put all the winnings from poker into then bankroll. At this writing, one of your authors plays professionally. As a result, he spends money on things like mortgage payments, food, and so on. We speak about "win rate" and "bankroll" as it relates to risk of ruin using the same terms we do to describe the rate at which we win in a particular game and the total amount of money we have on hand to play poker with. But these are not at all the same.

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First of all, in the United States, poker winnings are taxable. This can impact "win rate" by 30% or more, depending on how successful a player is. Second, "bankroll" should really be considered to be the amount of money we are willing to lose playing poker at a particular game and limit. For example, if a player plays $20-40 and has $12,000 set aside to play poker with, he might be said to have a bankroll of 300 bets. But if he will quit poker after losing $6,000, for example, or if he will drop down to a lower limit (a topic we will take up later), his effective bankroll for $20-40 is only 150 bets. He is not "ruined" in the sense of having no more capacity to play if he loses this amount, but it is illogical to consider the chance that he will lose $12,000 playing $20-40. In fact, he will never lose $12,000 at this game. His "true" risk of ruin is some combination of his risk of losing at each game the amount that he will tolerate losing until he will step down yet again in limits, or stop playing poker altogether.

Third, professional players have living expenses. These have to be met whether the player wins or loses. This represents a steady outflow of money that must be considered hi terms of bankroll and poker winnings. Even winning players who do not play professionally' usually do not leave 100% of then poker winnings in the bankroll. They instead spend the money on vacations, jewelry, or plasma TVs. There are some different reasonable strategies for extracting money from poker winnings to meet expenses, whether luxury or necessity, but your authors prefer a particular method which enables us to continue to rationally assess risk while making steady outflows of money. We simply pay ourselves a salary denominated in some time unit and reduce our win rate by that amount. This is quite similar to starting a small business enterprise.

This may raise eyebrows with some of our readers. *Deduct the salary from win rate! What if the amount of money I need is almost as much as I make?* Herein lies the rub. If a player's adjusted win rate (after deducting the money that the player takes out for expenses) is close to zero (or even negative!), then ***that player's risk of ruin is dramatically increased and will often be close to 100%.*** Countless pros have ignored this principle, skimmed most of their profit off of the top of their bankroll, and eventually gone bust when a severe negative swing hit. Even if a player doesn't go broke, he often must forego income by being forced to drop down in limits (we will have more to say on this topic later.)

Every dollar removed from a bankroll reduces that bankroll's resiliency to negative swings, and so when calculating risk of ruin, it is imperative to use a win rate that accurately characterizes ***not the amount of money won, but the amount of money put back into the bankroll as the game is played.***

For the aspiring professional, it gets worse; we have just shown a method by which we can assess the weighted risk of ruin, which is heavily impacted by the chance of a player's win rate being lower than his historical results would indicate. However, many players who are considering going professional after a somewhat longer time playing as an "amateur" have reached a threshold with their unadjusted win rate where the adjustment is not that large.

An example may help to elucidate this. Consider a player who has played 40.000 hands of his favorite game, and won 2.5 bets/100 hands with a standard deviation of 18 bets/100 hands. He has a bankroll of 300 bets. We can apply the RoRU function to find this player's weighted risk of ruin:

RoRU = 3.67%
RoR (simple) = 0.98%

The difference between these two numbers is just 2.69%; this is because the player has built up enough data that he is now almost three standard errors away from a zero win rate. As a result,

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his weighted risk of ruin does not suffer much from being subjected to uncertainty.

The player is now considering going professional. He plans to take out from his monthly win approximately 1 bet/100 hands he plays to pay his modest living expenses, and to allow his bankroll to grow with the remainder. We can now recalculate his RoRU and RoR simple:

RoRU = 17.25%
RoR (simple) = 6.22%

What has happened here is that dropping from 2.5 bets/100 hands to 1.5 bets/100 hands (by taking out 1 bet/100 hands for expenses) has brought his risk of ruin higher, but also ***significantly raised his chance of being a losing player after expenses.*** Here, he is less than two standard deviations above zero. Remember, it is not enough to be a winner overall: a player must be a winner, and a significant one, in terms of the money that is put back into his bankroll in order to have a risk of ruin below one. In the case of professional players, this includes fiving expenses, business expenses, and so on. It also includes positive flows, such as a pro player's salary or income from investments in other players.

Virtual Bankroll

This even applies to non-professional players who only take money out of the bankroll after winning streaks; doing this decreases the capacity of the bankroll to weather future downswings and increases the risk of ruin of the bankroll. But this brings up another important consideration, which is the concept of ***virtual bankroll.*** After all, we do not play in a complete vacuum; for non-professionals and even some professionals, there may be money available to flow into the bankroll from other jobs' salaries, investments, or other sources of income. To do an exhaustive analysis of poker risk of ruin, all of these factors would need to be considered. But in some cases, an adjustment made in broad strokes is sufficient. For example, a serious player who has a job that pays $300,000 a year has a virtual bankroll much larger than his present one, while a player who has no job at all and very few skills with which to get one might treat his poker bankroll a little more gingerly because it cannot be easily replaced.

The point of all this is that the impact on risk of ruin is not symmetric—slightly overestimating your win rate causes you underestimate your risk of ruin by a lot, while slightly underestimating it only causes you to overestimate it by a little.

One consideration that must color all of this analysis in a very important way is that game conditions are not static. Games ebb and flow, get better and worse. With the advent of televised poker, the world of poker players has grown immensely. It may be the case that this is a permanent change, but it's certainly not guaranteed. The same tiling is true of online poker. These types of things may also lead to an effect we discussed earlier. Upswings in profit are also correlated to favorable changes in conditions. Often when players are considering playing professionally, they have experienced some kind of winning streak. This may be due to a change in conditions in their preferred game which may or may not be permanent. The fact that conditions change makes it likely that most players ought to consider their overall risk of ruin to be slightly higher than the formulas presented here would suggest because it is more dramatically affected when conditions become worse than when they become better.

Key Concepts

- When we assess risk of ruin in the real world, we lack accuracy in our parameter estimates, such as win rate or standard deviation.
- We can partially correct for this by assuming that our true distribution of win rates is a normal distribution surrounding our point estimate.

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- The risk of ruin under uncertainty model is the resultant risk of ruin formula, which includes an adjustment for the uncertainty surrounding the win rate point estimate.
- This method can be applied to situations where only a small sample has been recorded, and includes the idea that the effect of error in win rate estimation is not symmetric.
- When considering becoming a professional, a realistic understanding of risk is an important and often neglected consideration.

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